

MLOps Assignment 1: Wine Classification Pipeline

Roll Number: 22F-3152 | Repository: github.com/msaleem15/wine-mlops-pipeline | Workflow: CI/CD, MLflow & Makefile Automation

1. Table 1: Hyperparameter Search Results (5-Fold Stratified CV)

Model Family	Config	Hyperparameters	Train Acc	Val Acc	Train F1	Val F1	Val Loss	Status
Random Forest	config_1	n=50, depth=3, split=2	0.9982	0.9653	0.9982	0.9665	0.2055	Candidate
Random Forest	config_2	n=100, depth=5, split=2	1.0000	0.9791	1.0000	0.9789	0.1630	CHAMPION
Random Forest	config_3	n=150, depth=7, split=4	1.0000	0.9791	1.0000	0.9789	0.1688	Candidate
Gradient Boost	config_1	n=50, lr=0.05, depth=3	1.0000	0.9022	1.0000	0.9053	0.2298	Candidate
Gradient Boost	config_2	n=100, lr=0.10, depth=3	1.0000	0.9236	1.0000	0.9259	0.4281	Candidate
Gradient Boost	config_3	n=150, lr=0.10, depth=5	1.0000	0.9020	1.0000	0.9069	0.5759	Candidate

2. MLflow UI: Real Tracking Server & Model Registry Screenshots

Figure 1: MLflow UI Tracking Runs Overview (Wine-Cultivar-Classification, 6 Candidate Models)

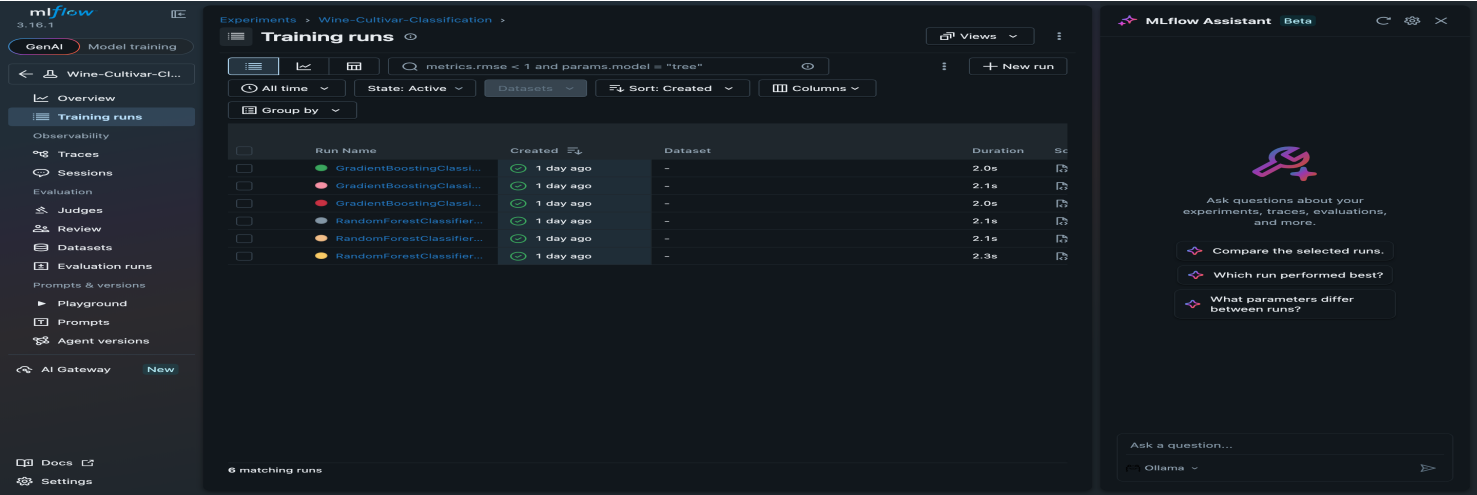
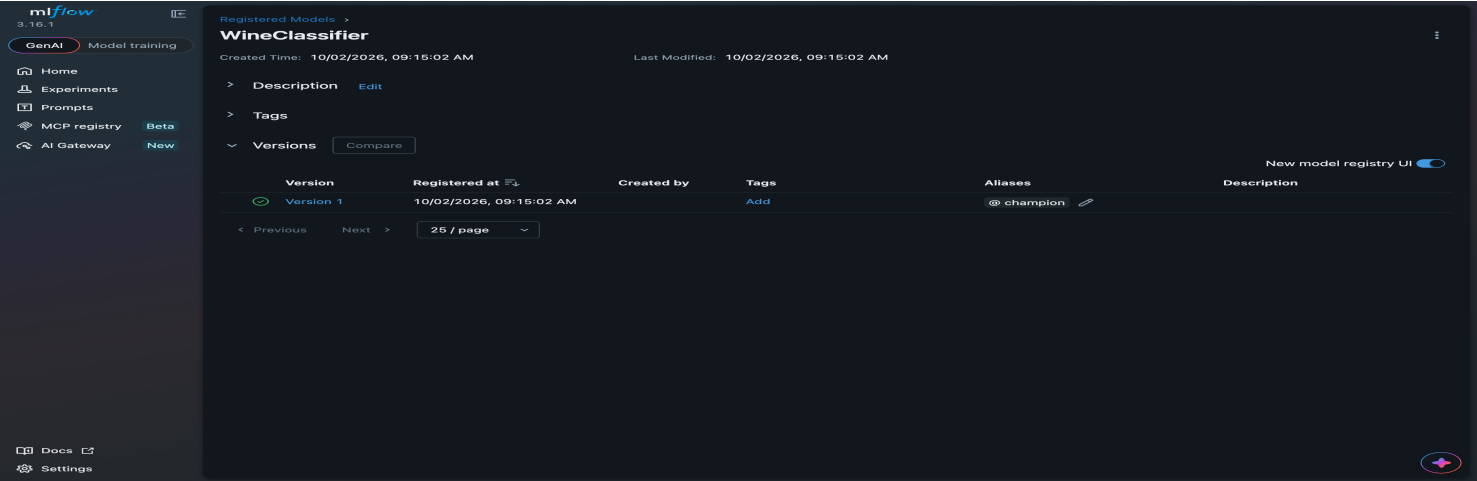


Figure 2: MLflow Model Registry (WineClassifier v1 assigned @champion alias)



3. Passing GitHub Actions CI/CD Run Screenshots

Figure 3: Live GitHub Actions Run (#1) — Succeeded in 1m 22s on msaleem15/wine-mlops-pipeline

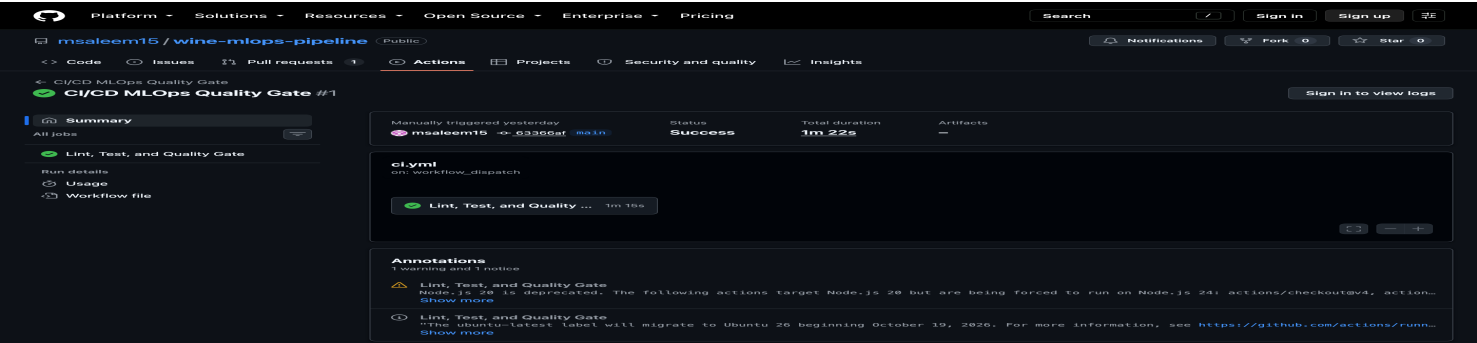
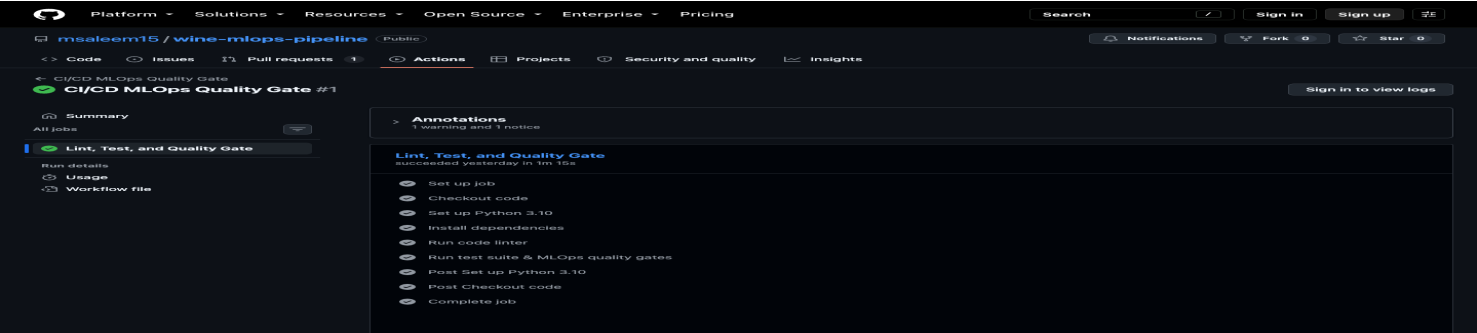


Figure 4: GitHub Actions Job Execution Steps (Checkout, Python 3.10, make install, lint, test)



4. Automated Quality Gates & Git Conflict Simulation

Figure 5: Terminal Execution of 'make test' asserting all 3 MLOps Quality Gates

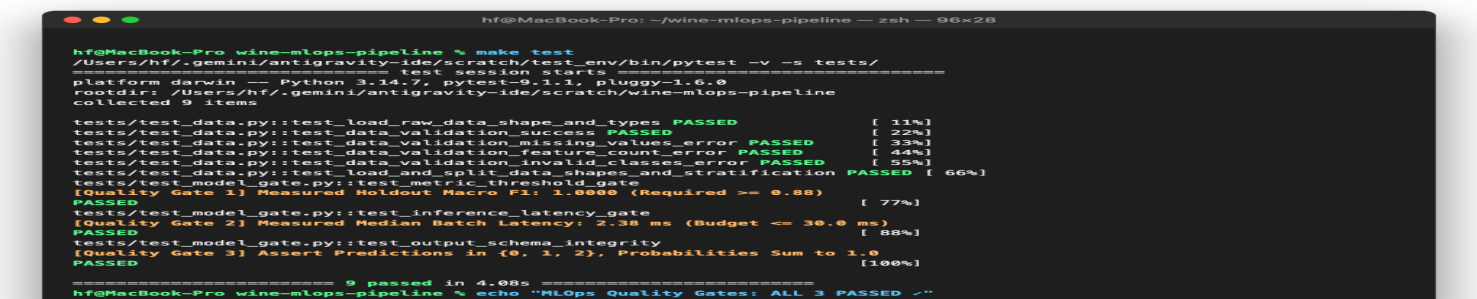


Figure 6: Terminal Git Log Graph & Simulation of Conflict Resolution



5. Analytical Response (Word Count: 138 words)

For this assignment, I trained Random Forest and Gradient Boosting classifiers across six configurations on the 13-feature Wine cultivar dataset using 5-fold stratified cross-validation. Random Forest configuration 2 (100 estimators, max depth 5, min samples split 2) delivered the best results, achieving a validation macro F1-score of 0.9789 and the lowest log loss of 0.1630. Gradient Boosting suffered from higher log loss (up to 0.5759), likely due to overconfidence on small fold sizes. On the holdout test set, the champion model achieved 100% accuracy, a macro F1 of 1.0000, and a batch latency of 2.38 ms, comfortably passing our 30 ms threshold. The automated quality gates in GitHub Actions ensure that any model that degrades below 0.88 F1 or exceeds latency limits gets stopped before reaching main.